International Journal of Research In Vocational Studies (IJRVOCAS)
Vol. 2 No. 4 (2023): IJRVOCAS – Special Issues – INCOSTIG – PP. 120~125
Print ISSN 2777-0168| Online ISSN 2777-0141| DOI prefix: 10.53893
https://journal.gpp.or.id/index.php/ijrvocas/index

Content-Based Filtering Recommendation System Using Categories Search Engine

Vicky Rolanda[1], Teddy Surya Gunawan[2] & Wanayumini[3]
[1,3]Department of Computer Engineering, Politeknik Negeri Medan, Indonesia
[2]Department of Computing, Standford University, USA

Keywords:

System recommendations
Content-based filtering
Search Recommendations
Category recommendations
Search by category

ABSTRACT

The search function is often used with certain keywords, usually using keywords and also catagories, someone knows what they want to search for, but some people often search according not to what they want to find but what is recommended to be seen based on ratings that can be by categories. Based on the recommendation system that is often applied, it is often developed based on the similarity between one item and another. So, this time the author wants to develop a search by users based on the same category so that users can reduce their desire to search according to the category they want, in this paper the author uses a recommendation system based on content base filtering.

Corresponding Author:

Vicky Rolanda,
Department of Computer Engineering and Information,
Politeknik Negeri Medan,
Almamater Road No 1, Padang Bulan, Medan, North Sumatera, Indonesia.
Email: vickyrolanda@gmail.com

1. INTRODUCTION

Information technology is very important where the development of digitalization is growing with information technology, the quality of products & services will continue to increase. In addition, users are the main factor in determining the success of a digital platform then we need a module that will help users to find what users want [1].

The search function in a platform is one of the most important things, besides helping users find what they are looking for, they can also provide recommendations according to the interests of the selected category, so to make this recommendation system I propose a method using content-base filtering.

Modern web applications or platforms that are currently widely used such as e-commerce, Netflix, youtube and sportify on average provide categories as a collection of items into a theme that will make it easier for users to find what they are interested in based on item categorization.

A recommendation system or recommendation system is a mechanism for providing personalized item recommendations to users that may suit the user's needs. sometimes the recommendation system also provides suggestions on the similarity of characters between one user and another like they both like the item [2].

There are three basic forms of recommendation systems mostly used on various platforms such as Content-based filtering, Collaborative filtering and Hybrid filtering [3]. The content-based filtering method recommends several objects based on the similarity of the recommended object to the selected object Content-based method is based on the content [4]. For example, if User A likes item 1, and item 2 is similar to item 1, then User A might also like item 2. Two similar products if they belong to the same category will be recommended to the user.

Similar research has been conducted by Resha Havilah Mondi, Ardhi Wiyajanto & Winarno by using content-based filtering to recommend information culinary arts by using user independence to recommend similar restaurants based on user content features[4].

From previous research, the author provides the option to provide recommendations based on search by category so as to reduce the scope of the search and increase the level of accuracy for recommending items, for example in the case of restaurants, if the user is given the choice of a Padang restaurant or a western restaurant, etc., the search will be more specific and increase accuracy. items to recommend. Likewise with e-commerce & many other digital platforms, adding a Content-base filtering method to search by category will increase the accuracy value of items that will be recommended to users.

2. RESEARCH METHOD

Content-based filtering methods are based on the assumption that there are multiple keywords associated with a certain item level. This can be observed from the system works that these keywords tend to detect and identify content related to it items. One can find many types of recommendations in the form of trying algorithms encourage various forms of recommendations based on specific items [3]. Recommend items relevant to specific users. You must first select a similarity metric (eg, product title, category etc).

Then, you must set up the system to rate each candidate item according to this similarity metric. Note that the recommendations are specific to this user, as the model does not use any information about other users.

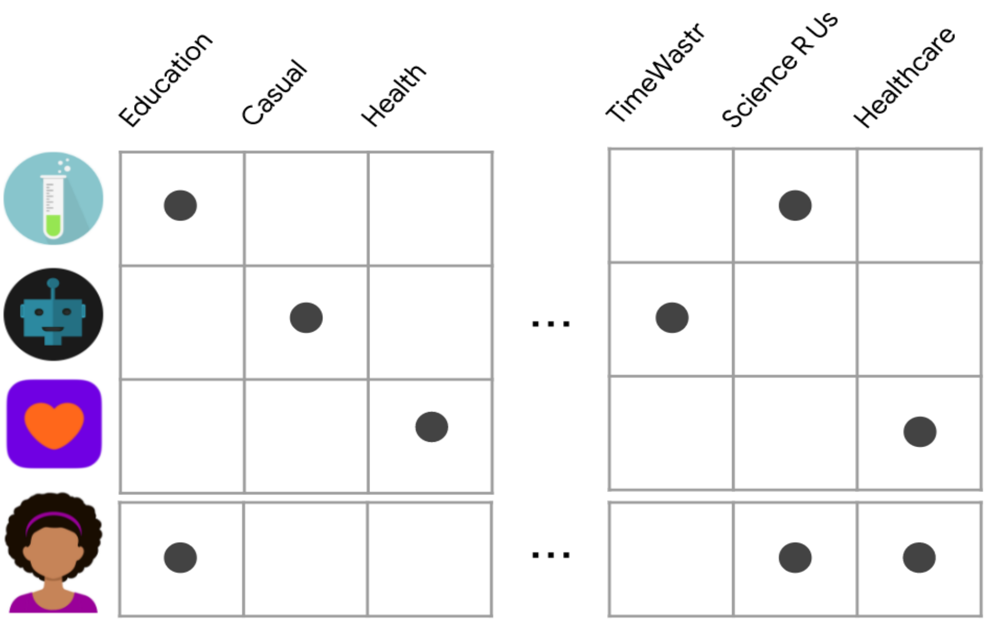

Figure 1 Ilustration Content-based
Source : (developers.google.com)

Using Dot product as a Similarity Measure Consider a case where a user embeds x and embedding y both are binary vectors. $\langle x, y \rangle = \sum_{i=1}^{d} x_i y_i$, features that appear in x and y contributes 1 to the amount. In other words, is the number of features that are active in both vectors simultaneously. The high point product then exhibits more common features, resulting in higher similarity [6].

https://doi.org/10.53893/ijrvocas.v2i4.177

The author wants to use a content-base filtering recommendation system workflow as shown in Figure 2 below:

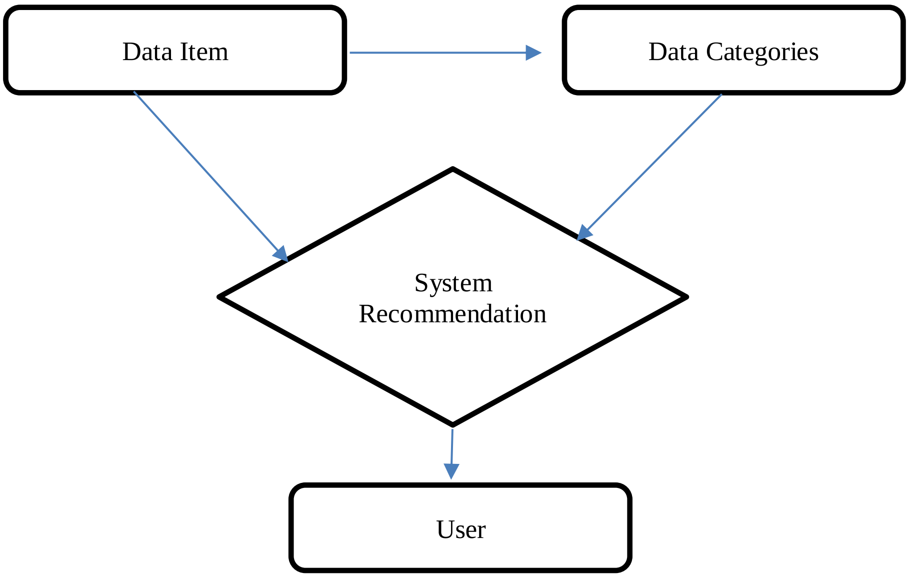

Figure 2 Content-based filtering

From Figure 2 Content-based Filltering starts from collecting data by having category attributes, after data collection, run Content-based filtering calculations then prediction results will be generated and recommended to users.

3. RESULTS AND ANALYSIS

In this paper obtained from the MovieLens dataset, the dataset used is a public dataset so that it can be used for the same test case with different methods. The MovieLens dataset contains thousands of movies rated by hundreds of users, created by GroupLens Research. This data set will also always change over time, and if in the future there are studies related to the same dataset, it is possible that the dataset may also vary in the content of data values. This research uses a dataset of 100,836 ratings and 3,683 tag applications in 9,742 films. This data was created by 610 users between March 29, 1996 and September 24, 2018.

```python
import warnings
warnings.filterwarnings("ignore")

## for data
import pandas as pd
import numpy as np
import re
from datetime import datetime

## for plotting
import matplotlib.pyplot as plt
import seaborn as sns

## for machine learning
from sklearn import metrics, preprocessing

## for deep learning
from tensorflow.keras import models, layers, utils  #(2.6.0)
```

Figure 4 Import Module

In Figure 4 the Import Module is used to enter the modules used in the test cases. As Panda is used to analyze data, Numpy is used to provide support for large multidimensional sets and matrices.

	movieId	title	genres	movie	name	date	old
0	1	Toy Story (1995)	Adventure|Animation|Children|Comedy|Fantasy	0	Toy Story	1995	1
1	2	Grumpier Old Men (1995)	Comedy|Romance	1	Grumpier Old Men	1995	1
2	3	Waiting to Exhale (1995)	Comedy|Drama|Romance	2	Waiting to Exhale	1995	1
3	4	Father of the Bride Part II (1995)	Comedy	3	Father of the Bride Part II	1995	1
4	5	Sabrina (1995)	Comedy|Romance	4	Sabrina	1995	1
...	...	...	...	...	...	...	...
3751	3752	Gintama: The Movie (2010)	Action|Animation|Comedy|Sci-Fi	3751	Gintama: The Movie	2010	0
3752	3753	Silver Spoon (2014)	Comedy|Drama	3752	Silver Spoon	2014	0
3753	3754	Black Butler: Book of the Atlantic (2017)	Action|Animation|Comedy|Fantasy	3753	Black Butler: Book of the Atlantic	2017	0
3754	3755	No Game No Life: Zero (2017)	Animation|Comedy|Fantasy	3754	No Game No Life: Zero	2017	0
3755	3756	Andrew Dice Clay: Dice Rules (1991)	Comedy	3755	Andrew Dice Clay: Dice Rules	1991	1

3756 rows × 7 columns

Figure 5 Dataset

Figure 5 The dataset is a table of movies, each row represents an item and the two columns on the right contain the genre, which is static (tags can see it as movie metadata).

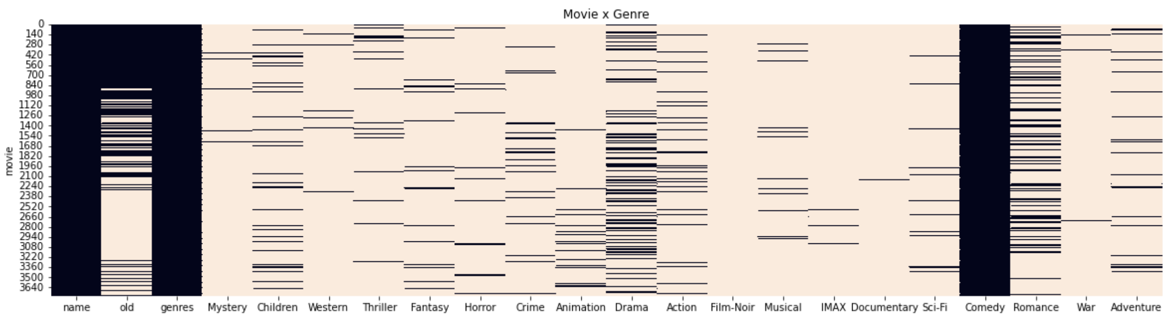

Figure 6 Content-based filtering

The graph above can be seen from the matrix which is less frequent because most films do not have all genres, in writing this journal the author chooses the Comedy category to be the test data.

https://doi.org/10.53893/ijrvocas.v2i4.177

```
# Model
usr = train[["y"]].fillna(0).values.T
prd = dtf_products.drop(["name","genres"],axis=1).values
print("Pengguna", usr.shape, " x  Movies", prd.shape)
```

```
Pengguna (1, 3756)  x  Movies (3756, 20)
```

Figure 7 User Model & Movie

From the data generated, it can be seen that there is 1 user with 3756 films and 3576 films with 20 different genres then the weights must be re-applied to the Item-Feature matrix to get the predicted rating.

```
print("--- Pengguna", i, "---")

top = 5
y_test = test.sort_values("y", ascending=False)["movie"].values[:top]
print("y_test:", y_test)

predicted = test.sort_values("yhat", ascending=False)["movie"].values[:top]
print("prediksi:", predicted)

true_positive = len(list(set(y_test) & set(predicted)))
print("jumlah & presentasi:", true_positive, "("+str(round(true_positive/top*100,1))+"%)")
print("akurasi:", str(round(metrics.accuracy_score(y_test,predicted)*100,1))+"%")
print("rating rata - rata:", round(mean_reciprocal_rank(y_test, predicted),2))
```

```
--- Pengguna 2 ---
y_test: [3023. 3209. 3702.]
prediksi: [3702. 3023. 3209.]
jumlah & presentasi: 3 (60.0%)
akurasi: 0.0%
rating rata - rata: 0.61
```

Figure 8 Rersult Recommendation System

From the number of test data [3023 3209 3702] users have predictions [3702 3023 3209] with the recommended amount being 3 (60%) of the total number of past tests with an average rating of 0.61.

	movie	y	yhat	name	old	genres
698	3702.0	0.5	0.433030	The Nut Job 2: Nutty by Nature	0	Adventure|Animation|Children|Comedy
19	3023.0	0.5	0.424361	Angels' Share, The	0	Comedy|Drama
205	3209.0	0.5	0.333333	Honest Liar, An	0	Comedy|Documentary

Figure 9 Movie Recommendation System List.

Figure 9 menunjukan daftar film yang di rekomendasikan dengan kategori comedy dengan yhat tertinggi 0.43 menunjukan hasil rating tertinggi dari beberapa data test lainnya.

4. CONCLUSION

Search recommendation data by category is highly ordered for many cases of digital platforms, based on the test results above, recommendation data based on genre and film rating can be generated to provide recommendations to users. Thus, users will be more helpful in determining the film they want to see every time they search by genre, new films will be recommended based on the film's rating level.

ACKNOWLEDGEMENTS

I thank you for the sources I have quoted to complete this journal, I hope this journal can be used as a learning material and a comparison of content-base filtering methods with other recommended system methods.

REFERENCES

[1] R. Roslina, S. Saulan, and Wanayumini, ‘Model of Information Technology Facility Service Based on user Satisfaction’, *J. Phys. Conf. Ser.*, vol. 1361, no. 1, 2019, doi: 10.1088/1742-6596/1361/1/012032.
[2] W. M. Azmy and O. Emam, ‘Content-Based Recommendation System Using Search Engine’, no. November, 2016.
[3] J. C. Arrias, D. Alvarado, and M. Calderón, ‘No 主観的健康感を中心とした在宅高齢者における 健康関連指標に関する共分散構造分析Title’, pp. 5–10, 2019.
[4] R. H. Mondi, A. Wijayanto, and Winarno, ‘Recommendation System With Content-Based Filtering Method for Culinary Tourism in Mangan Application’, *J. Ilm. Teknol. dan Inf.*, vol. 8, no. 2, pp. 65–72, 2019, [Online]. Available: https://jurnal.uns.ac.id/itsmart/article/download/35008/27748.
[5] J. W. G. Putra, *Pengenalan Konsep Pembelajaran Mesin dan Deep Learning*. 2020.
[6] developers.google.com, ‘Content-based Filtering’, 2021, [Online]. Available: https://developers.google.com/machine-learning/recommendation/content-based/basics.

How to Cite

Rolanda, V., Gunawan, T. S., & Wanayumini. (2023). Content-Based Filtering Recommendation System Using Categories Search Engine. International Journal of Research in Vocational Studies (IJRVOCAS), 2(4), 120–125. https://doi.org/10.53893/ijrvocas.v2i4.177